Lecture 2-9 — Form Events & Validation

This note shows how to handle form submission, validate with built-in browser checks, and read form data with `FormData`.

User Registration

Full Name:

Email:

Age:

Gender:

Male

Female

Skills:

JavaScript

HTML

CSS

Submit

Output:

No submissions yet.

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How it works (beginners)

- Listen for the `submit` event and call `preventDefault()` to stop automatic page reload.
- Run `checkValidity()` and `reportValidity()` to let the browser show validation UI.
- Use `FormData(form)` to read all form fields easily, including checkboxes and files.

Advanced Form Notes

Multiple fields with same name

When several inputs share the same name (e.g., checkboxes), `FormData` will provide multiple entries for that name. To collect them as an array:

```
const fd = new FormData(form);
const skills = [];
for(const [k,v] of fd.entries()){
  if(k === 'skills') skills.push(v);
}
// now skills is an array of selected values
```

Files

File inputs are also returned in FormData as File objects. You can send FormData directly in a `fetch` request without manual serialization.

Custom validation

Use `setCustomValidity` to provide custom messages and control validity:

```
const name = form.querySelector('[name="fullname"]');
if(name.value.length < 3){
  name.setCustomValidity('Name must be at least 3 characters');
} else {
  name.setCustomValidity('');
}
if(!form.checkValidity()) form.reportValidity();
```


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FormData → JSON example

If you need JSON (for an API that doesn't accept multipart), convert FormData:

```
const obj = {};
for(const [k,v] of fd.entries()){
  if(obj[k]){
    // convert to array when multiple values
    obj[k] = Array.isArray(obj[k]) ? obj[k].concat(v) : [obj[k], v];
  } else obj[k] = v;
}
const json = JSON.stringify(obj);
```

Example: submit via fetch

```
// send FormData directly
fetch('/submit', { method: 'POST', body: fd });

// or send JSON
fetch('/submit-json', { method: 'POST', headers: { 'Content-
Type':'application/json' }, body: json });
```

Exercise

Extend the form to include a file input and show selected filenames in the output box. Then try sending the FormData to a dummy server (or inspect network requests in DevTools).

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